About the Data

In this notebook, we will be working with 2 data sets:

- Facebook's stock price throughout 2018 (obtained using the stock_analysis package).
- daily weather data for NYC from the National Centers for Environmental Information (NCEI) API.

Note: The NCEI is part of the National Oceanic and Atmospheric Administration (NOAA) and, as you can see from the URL for the API, this resource was created when the NCEI was called the NCDC. Should the URL for this resource change in the future, you can search for the NCEI weather API to find the updated one.

Background on the weather data

Data meanings: AWND : average wind speed PRCP : precipitation in millimeters SNOW : snowfall in millimeters SNWD : snow depth in millimeters TMAX : maximum daily temperature in Celsius TMIN : minimum daily temperature in Celsius

Setup

```
import numpy as np
import pandas as pd

weather = pd.read_csv('nyc_weather_2018.csv',parse_dates=['date'])
weather.head()
```

```
        date datatype            station attributes  value
0 2018-01-01     PRCP  GHCND:US1CTFR0039   ,,N,0800    0.0
1 2018-01-01     PRCP  GHCND:US1NJBG0015   ,,N,1050    0.0
2 2018-01-01     SNOW  GHCND:US1NJBG0015   ,,N,1050    0.0
3 2018-01-01     PRCP  GHCND:US1NJBG0017   ,,N,0920    0.0
4 2018-01-01     SNOW  GHCND:US1NJBG0017   ,,N,0920    0.0
```

```
fb = pd.read_csv('fb_2018.csv', index_col='date', parse_dates=True)
fb.head()
```

```
              open    high       low   close    volume
date
2018-01-02  177.68  181.58  177.5500  181.42  18151903
2018-01-03  181.88  184.78  181.3300  184.67  16886563
2018-01-04  184.90  186.21  184.0996  184.33  13880896
2018-01-05  185.59  186.90  184.9300  186.85  13574535
2018-01-08  187.20  188.90  186.3300  188.28  17994726
```

Arithmetic and statistics

We already saw that we can use mathematical operators like + and / with dataframes directly. However, we can also use methods, which allow us to specify the axis to perform the calculation over. By default this is per column. Let's find the z-scores for the volume traded and look at the days where this was more than 3 standard deviations from the mean:

```
fb.assign(abs_z_score_vloume=lambda x:
x.volume.sub(x.volume.mean()).div(x.volume.std()).abs()).query('abs_z_
score_vloume>3')
```

```
              open    high     low   close     volume
abs_z_score_vloume
date

2018-03-19  177.01  177.17  170.06  172.56   88140060
3.145078
2018-03-20  167.47  170.20  161.95  168.15  129851768
5.315169
2018-03-21  164.80  173.40  163.30  169.39  106598834
4.105413
2018-03-26  160.82  161.10  149.02  160.06  126116634
5.120845
2018-07-26  174.89  180.13  173.75  176.26  169803668
7.393705
```

We can use rank() and pct_change() to see which days had the largest change in volume traded from the day before:

```
fb.assign(volume_pct_change=fb.volume.pct_change(),
          pct_change_rank=lambda x:
x.volume_pct_change.abs().rank(ascending=False)).nsmallest(5,
'pct_change_rank')
```

```
              open    high     low   close     volume
volume_pct_change  \
date

2018-01-12  178.06  181.48  177.40  179.37   77551299
7.087876
2018-03-19  177.01  177.17  170.06  172.56   88140060
2.611789
2018-07-26  174.89  180.13  173.75  176.26  169803668
1.628841
2018-09-21  166.64  167.25  162.81  162.93   45994800
1.428956
2018-03-26  160.82  161.10  149.02  160.06  126116634
1.352496
```

```
            pct_change_rank
date
2018-01-12              1.0
2018-03-19              2.0
2018-07-26              3.0
2018-09-21              4.0
2018-03-26              5.0
```

January 12th was when the news that Facebook changed its news feed product to focus more on content from a users' friends over the brands they follow. Given that Facebook's advertising is a key component of its business (nearly 89% in 2017), many shares were sold and the price dropped in panic:

```
fb['2018-01-11':"2018-01-12"]
```

```
               open    high     low   close    volume
date
2018-01-11  188.40  188.40  187.38  187.77   9588587
2018-01-12  178.06  181.48  177.40  179.37  77551299
```

Throughout 2018, Facebook's stock price never had a low above $215:

```
(fb > 215).any()
```

```
open       True
high       True
low       False
close      True
volume     True
dtype: bool
```

```
(fb > 215).all()
```

```
open      False
high      False
low       False
close     False
volume     True
dtype: bool
```

Binning and thresholds

When working with the volume traded, we may be interested in ranges of volume rather than the exact values. No two days have the same volume traded:

```
(fb.volume.value_counts() > 1).sum()
```

```
np.int64(0)
```

We can use pd.cut() to create 3 bins of even an even range in volume traded and name them. Then we can work with low, medium, and high volume traded categories:

```
volume_binned = pd.cut(fb.volume, bins=3, labels=['low', 'med',
'high'])
volume_binned.value_counts()
```

```
volume
low     240
med       8
high      3
Name: count, dtype: int64
```

```
fb[volume_binned == 'high'].sort_values(
    'volume', ascending=False)
```

```
              open    high     low   close     volume
date
2018-07-26  174.89  180.13  173.75  176.26  169803668
2018-03-20  167.47  170.20  161.95  168.15  129851768
2018-03-26  160.82  161.10  149.02  160.06  126116634
```

July 25th Facebook announced disappointing user growth and the stock tanked in the after hours:

```
fb['2018-07-25':'2018-07-26']
```

```
               open    high     low   close     volume
date
2018-07-25  215.715  218.62  214.27  217.50   64592585
2018-07-26  174.890  180.13  173.75  176.26  169803668
```

Cambridge Analytica scandal broke on Saturday March 17th, so we look to the Monday for the numbers:

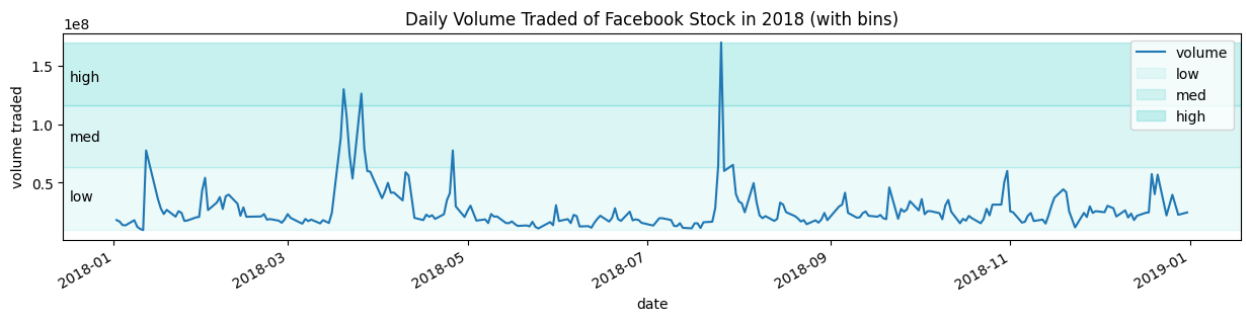
```
fb['2018-03-16':'2018-03-20']
```

```
              open    high     low   close     volume
date
2018-03-16  184.49  185.33  183.41  185.09   24403438
2018-03-19  177.01  177.17  170.06  172.56   88140060
2018-03-20  167.47  170.20  161.95  168.15  129851768
```

Since most days have similar volume, but a few are very large, we have very wide bins. Most of the data is in the low bin. Note: visualizations will be covered in chapters 5 and 6.

```
import matplotlib.pyplot as plt

fb.plot(y='volume', figsize=(15, 3), title='Daily Volume Traded of
Facebook Stock in 2018 (with bins)')
```

```
for bin_name, alpha, bounds in zip(
    ['low', 'med', 'high'], [0.1, 0.2, 0.3], pd.cut(fb.volume,
bins=3).unique().categories.values):
  plt.axhspan(bounds.left, bounds.right, alpha=alpha, label=bin_name,
color='mediumturquoise')
  plt.annotate(bin_name, xy=('2017-12-17', (bounds.left +
bounds.right)/2.1))


plt.ylabel('volume traded')
plt.legend()
plt.show()
```



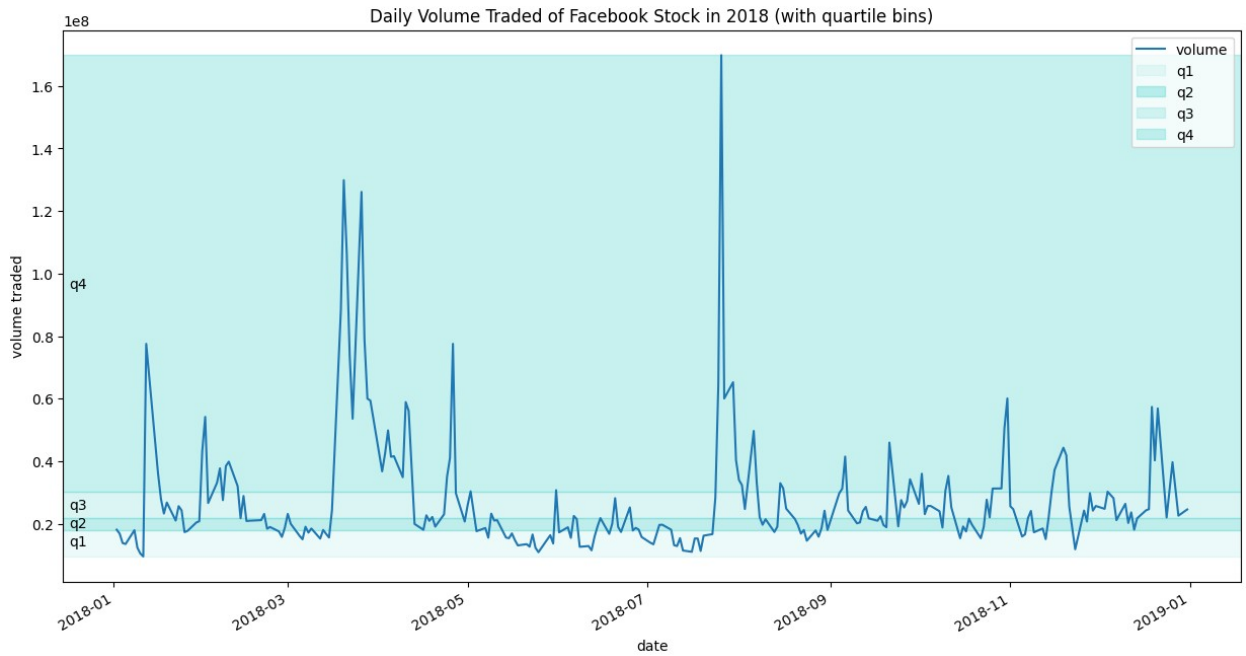
If we split using quantiles, the bins will have roughly the same number of observations. For this, we use qcut() . We will make 4 quartiles:

```
volume_qbinned = pd.qcut(fb.volume, q=4, labels=['q1', 'q2', 'q3',
'q4'])
volume_qbinned.value_counts()
```

```
volume
q1    63
q2    63
q4    63
q3    62
Name: count, dtype: int64
```

```
fb.plot(y='volume', figsize=(15, 8), title='Daily Volume Traded of
Facebook Stock in 2018 (with quartile bins)')

for bin_name, alpha, bounds in zip(
    ['q1', 'q2', 'q3', 'q4'], [0.1, 0.35, 0.2, 0.3],
pd.qcut(fb.volume, q=4).unique().categories.values):
  plt.axhspan(bounds.left, bounds.right, alpha=alpha, label=bin_name,
color='mediumturquoise')
  plt.annotate(bin_name, xy=('2017-12-17', (bounds.left +
bounds.right)/2.1))

plt.ylabel('volume traded')
```

```
plt.legend()
plt.show()
```



Sometimes we don't want to make bins, but rather cap values at a threshold. Before we look at an example, let's pivot our weather data for the Central Park station:

```
central_park_weather = weather.query(
    'station == "GHCND:USW00094728"'
).pivot(index='date', columns='datatype', values='value')
central_park_weather.head()

datatype     ADPT     ASLP     ASTP   AWBT  AWND  PRCP  RHAV  RHMN
RHMX  \
date

2018-01-01 -194.0  10278.0  10224.0 -122.0   3.5   0.0  48.0  34.0
60.0
2018-01-02 -156.0  10278.0  10227.0  -94.0   3.6   0.0  52.0  42.0
62.0
2018-01-03 -161.0  10237.0  10196.0  -78.0   1.4   0.0  42.0  28.0
51.0
2018-02-01  -39.0  10139.0  10085.0   11.0   1.7   1.3  56.0  45.0
76.0
2018-02-02 -117.0  10176.0  10105.0  -50.0   4.1   4.6  52.0  38.0
89.0

datatype    SNOW  ...  TMAX  TMIN   WDF2   WDF5  WSF2  WSF5  WT01
WT02  WT03  \
date              ...
```

```
2018-01-01   0.0  ...  -7.1 -13.8  300.0  300.0   6.7  11.2   NaN
NaN   NaN
2018-01-02   0.0  ...  -3.2 -10.5  260.0  250.0   7.2  12.5   NaN
NaN   NaN
2018-01-03   0.0  ...  -1.0  -8.8  260.0  270.0   6.3   9.8   NaN
NaN   NaN
2018-02-01   0.0  ...   7.2   0.0  250.0  250.0   5.8   9.4   NaN
NaN   NaN
2018-02-02   0.0  ...   5.0  -8.2  290.0  280.0   8.9  14.8   1.0
NaN   NaN

datatype    WT08
date
2018-01-01   NaN
2018-01-02   NaN
2018-01-03   NaN
2018-02-01   NaN
2018-02-02   NaN

[5 rows x 21 columns]
```

Say we don't care how much snow their was, just that it snowed in Central Park. However, we don't want to make a Boolean column since we need to preserve the data type of float. We can use clip() to replace values above a upper threshold with the threshold and replace values below a lower threshold with the lower threshold. This means we can use clip(0, 1) to change all the snow values of one or more to 1, which easily shows us the days snow was recorded in Central Park. Preserving the data type will save some work later on if we are building a model:

```
central_park_weather.SNOW.clip(0, 1).value_counts()

SNOW
0.0    43
1.0     1
Name: count, dtype: int64
```

Note: the clip() method can also be called on the dataframe itself.

Applying Functions

We can use the apply() method to run the same operation on all columns (or rows) of the dataframe. Let's calculate the z-scores of the TMIN, TMAX, and PRCP observations in Central Park in October 2018:

```
oct_weather_z_scores = central_park_weather.loc[
'2018-10', ['TMIN', 'TMAX', 'PRCP']].apply(lambda x:
x.sub(x.mean()).div(x.std()))
oct_weather_z_scores.describe().T
```

```
          count          mean  std       min       25%       50%
75%  \
datatype

TMIN        3.0  0.000000e+00  1.0 -0.577350 -0.577350 -0.577350
0.288675
TMAX        3.0 -1.332268e-15  1.0 -1.082543 -0.444616  0.193311
0.541271
PRCP        4.0  4.163336e-17  1.0 -0.538024 -0.538024 -0.479859
0.058165

               max
datatype
TMIN      1.154701
TMAX      0.889231
PRCP      1.497743
```

October 27th rained much more than the rest of the days:

```
oct_weather_z_scores.query('PRCP > 3')

Empty DataFrame
Columns: [TMIN, TMAX, PRCP]
Index: []
```

Indeed, this day was much higher than the rest:

```
central_park_weather.loc['2018-10', 'PRCP'].describe()

count     4.000000
mean      4.625000
std       8.596269
min       0.000000
25%       0.000000
50%       0.500000
75%       5.125000
max      17.500000
Name: PRCP, dtype: float64
```

When the function we want to apply isn't vectorized, we can:

- use np.vectorize() to vectorize it (similar to how map() works) and then use it with apply()
- use applymap() and pass it the non-vectorized function directly

Say we wanted to count the digits of the whole numbers for the Facebook data. len() is not vectorized:

```
import numpy as np
```

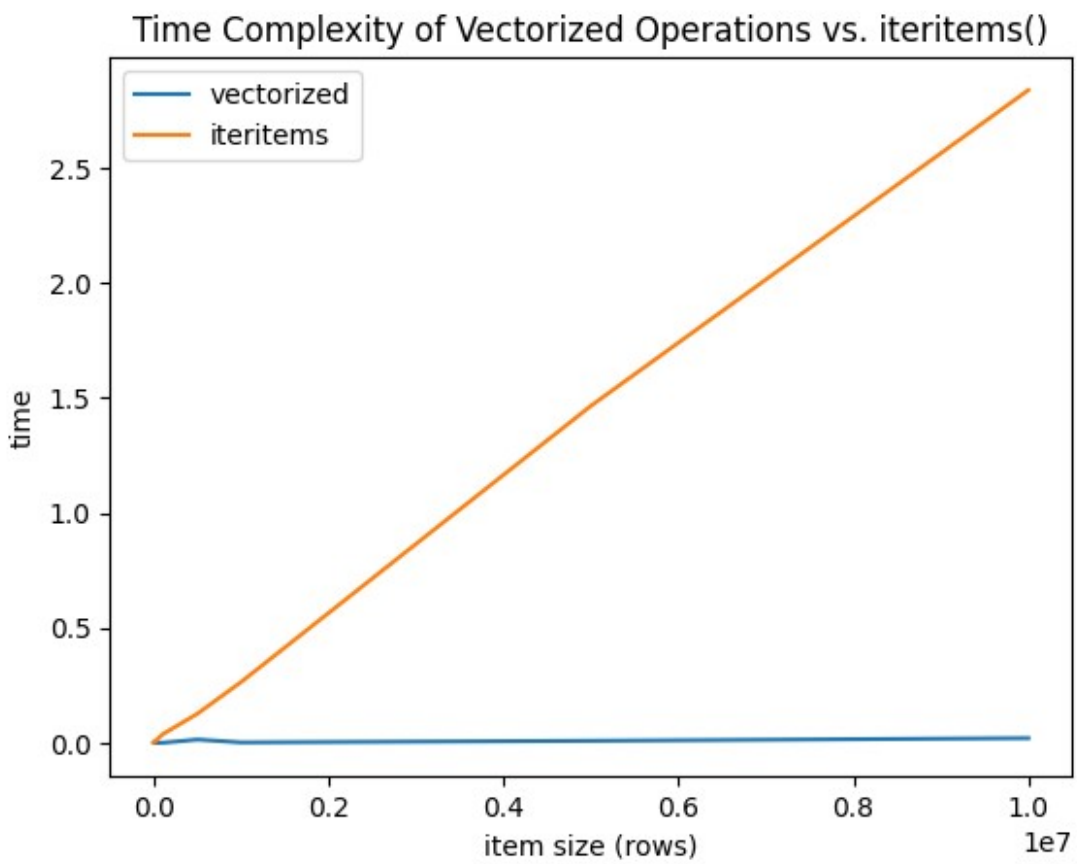
```
fb.apply(lambda x: np.vectorize(lambda y: len(str(np.ceil(y))))
(x)).astype('int64').equals(fb.applymap(lambda x:
len(str(np.ceil(x)))))
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_20084\2279867799.py in ?()
      1 import numpy as np
      2
----> 3 fb.apply(lambda x: np.vectorize(lambda y:
len(str(np.ceil(y))))(x)).astype('int64').equals(fb.applymap(lambda x:
len(str(np.ceil(x)))))

e:\Python\Lib\site-packages\pandas\core\generic.py in ?(self, name)
   6202             and name not in self._accessors
   6203             and
self._info_axis._can_hold_identifiers_and_holds_name(name)
   6204         ):
   6205             return self[name]
-> 6206         return object.__getattribute__(self, name)

AttributeError: 'DataFrame' object has no attribute 'applymap'
```

A simple operation of addition to each element in a series grows linearly in time complexity when using iteritems() , but stays near 0 when using vectorized operations. iteritems() and related methods should only be used if there is no vectorized solution:

```
import time
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

np.random.seed(0)
vectorized_results = {}
iteritems_results = {}

for size in [10, 100, 1000, 10000, 100000, 500000, 1000000, 5000000,
10000000]:
  test = pd.Series(np.random.uniform(size=size))
  start = time.time()
  x = test + 10
  end = time.time()
  vectorized_results[size] = end - start

  start = time.time()
  x = []
  for i, v in test.items():
    x.append(v + 10)
```

```
  x = pd.Series(x)
  end = time.time()
  iteritems_results[size] = end - start

pd.DataFrame([pd.Series(vectorized_results, name='vectorized'),
pd.Series(iteritems_results, name='iteritems')]
            ).T.plot(title='Time Complexity of Vectorized Operations
vs. iteritems()')
plt.xlabel('item size (rows)')
plt.ylabel('time')
plt.show()
```



Window Calculations

Consult the understanding windows calculation notebook for interactive visualizations to help understand window calculations.

The rolling() method allows us to perform rolling window calculations. We simply specify the window size (3 days here) and follow it with a call to an aggregation function (sum here):

```
central_park_weather.loc['2018-10'].assign(rolling_PRCP=lambda x:
x.PRCP.rolling('3D').sum())[['PRCP', 'rolling_PRCP']].head(7).T
```

```
date          2018-10-01  2018-10-02  2018-10-03  2018-10-04
datatype
PRCP                 0.0        17.5         0.0         1.0
rolling_PRCP         0.0        17.5        17.5        18.5
```

We can also perform the rolling calculations on the entire dataframe at once. This will apply the same aggregation function to each column:

```
central_park_weather.loc['2018-
10'].rolling('3D').mean().head(7).iloc[:, :6]

datatype          ADPT          ASLP          ASTP   AWBT      AWND
PRCP
date

2018-10-01  172.000000  10247.000000  10200.000000  189.0  0.900000
0.000000
2018-10-02  180.500000  10221.500000  10176.000000  194.5  0.900000
8.750000
2018-10-03  172.333333  10205.333333  10159.000000  187.0  0.966667
5.833333
2018-10-04  176.000000  10175.000000  10128.333333  187.0  0.800000
6.166667
```

We can use different aggregation functions per column if we use agg() instead. We pass in a dictionary mapping the column to the aggregation to perform on it:

```
central_park_weather['2018-10-01':'2018-10-07'].rolling('3D').agg(
    {'TMAX': 'max', 'TMIN': 'min', 'AWND': 'mean', 'PRCP': 'sum'}
).join( # join with original data for comparison
central_park_weather[['TMAX', 'TMIN', 'AWND', 'PRCP']],
    lsuffix='_rolling'
).sort_index(axis=1) # sort columns so rolling calcs are next to
originals

datatype    AWND  AWND_rolling  PRCP  PRCP_rolling  TMAX  TMAX_rolling
TMIN  \
date

2018-10-01   0.9      0.900000   0.0           0.0  24.4          24.4
17.2
2018-10-02   0.9      0.900000  17.5          17.5  25.0          25.0
18.3
2018-10-03   1.1      0.966667   0.0          17.5  23.3          25.0
17.2
2018-10-04   0.4      0.800000   1.0          18.5   NaN          25.0
NaN

datatype    TMIN_rolling
```

```
date
2018-10-01          17.2
2018-10-02          17.2
2018-10-03          17.2
2018-10-04          17.2
```

Rolling calculations (rolling()) use a sliding window. Expanding calculations (expanding()) however grow in size. These are equivalent to cumulative aggregations like cumsum() ; however, we can specify the minimum number of periods required to start calculating (default is 1):

```
central_park_weather.PRCP.expanding().sum().equals(central_park_weathe
r.PRCP.cumsum())

False
```

Separate expanding aggregations per column. Note that agg() will accept numpy functions too:

```
central_park_weather['2018-10-01':'2018-10-
07'].expanding().agg({'TMAX': 'max',

'TMIN': 'min',

'AWND': 'mean',

'PRCP': 'sum'}).join(central_park_weather[['TMAX', 'TMIN', 'AWND',
'PRCP']],
 lsuffix='_expanding'
).sort_index(axis=1)

datatype    AWND  AWND_expanding  PRCP  PRCP_expanding  TMAX
TMAX_expanding  \
date

2018-10-01   0.9        0.900000   0.0             0.0  24.4
24.4
2018-10-02   0.9        0.900000  17.5            17.5  25.0
25.0
2018-10-03   1.1        0.966667   0.0            17.5  23.3
25.0
2018-10-04   0.4        0.825000   1.0            18.5   NaN
25.0

datatype    TMIN  TMIN_expanding
date
2018-10-01  17.2            17.2
2018-10-02  18.3            17.2
2018-10-03  17.2            17.2
2018-10-04   NaN            17.2
```

We can calculate the exponentially weighted moving average as follows. Note that span here is the periods to use:

```
fb.assign(
 close_ewma=lambda x: x.close.ewm(span=5).mean()
).tail(10)[['close', 'close_ewma']]
```

```
              close  close_ewma
date
2018-12-17  140.19  142.235433
2018-12-18  143.66  142.710289
2018-12-19  133.24  139.553526
2018-12-20  133.40  137.502350
2018-12-21  124.95  133.318234
2018-12-24  124.06  130.232156
2018-12-26  134.18  131.548104
2018-12-27  134.52  132.538736
2018-12-28  133.20  132.759157
2018-12-31  131.09  132.202772
```

Consult the understanding_window_calculations.ipynb notebook for interactive visualizations to help understand window calculations.

Pipes

Pipes all use to apply any function that accepts our data as the first argument and pass in any additional arguments. This makes it easy to chain steps together regardless of if they are methods or functions:

We can pass any function that will accept the caller of pipe() as the first argument:

```
def get_info(df):
    return '%d rows and %d columns and max closing z-score was %d' %
(*df.shape, df.close.max())

fb.loc['2018-Q1'].apply(lambda x: (x - x.mean()) /
x.std()).pipe(get_info) == get_info(fb.loc['2018-Q1'].apply(lambda x:
(x - x.mean()) / x.std()))
```

```
True
```

For example, passing pd.DataFrame.rolling to pipe() is equivalent to calling rolling() directly on the dataframe, except we have more flexiblity to change this:

```
fb.pipe(pd.DataFrame.rolling,
'20D').mean().equals(fb.rolling('20D').mean())
```

```
True
```

The pipe takes the function passed in and calls it with the object that called pipe() as the first argument. Positional and keyword arguments are passed down:

```
pd.DataFrame.rolling(fb,
'20D').mean().equals(fb.rolling('20D').mean())

True
```

We can use a pipe to make a function that we can use for all our window calculation needs:

```
def window_calc(df, func, agg_dict, *args, **kwargs):
  return df.pipe(func, *args, **kwargs).agg(agg_dict)
```

We can use the same interface to calculate various window calculations now. Let's find the expanding median for the Facebook data:

```
window_calc(fb, pd.DataFrame.expanding, np.median).head()

               open     high       low    close      volume
date
2018-01-02  177.68  181.580  177.5500  181.420  18151903.0
2018-01-03  179.78  183.180  179.4400  183.045  17519233.0
2018-01-04  181.88  184.780  181.3300  184.330  16886563.0
2018-01-05  183.39  185.495  182.7148  184.500  15383729.5
2018-01-08  184.90  186.210  184.0996  184.670  16886563.0
```

Using the exponentially weighted moving average requires we pass in a keyword argument

```
window_calc(fb, pd.DataFrame.ewm, 'mean', span=3).head()

                   open        high         low       close
volume
date

2018-01-02  177.680000  181.580000  177.550000  181.420000
1.815190e+07
2018-01-03  180.480000  183.713333  180.070000  183.586667
1.730834e+07
2018-01-04  183.005714  185.140000  182.372629  184.011429
1.534980e+07
2018-01-05  184.384000  186.078667  183.736560  185.525333
1.440299e+07
2018-01-08  185.837419  187.534839  185.075110  186.947097
1.625679e+07
```

With rolling calculations, we can pass in a positional argument for the window size:

```
window_calc(central_park_weather.loc['2018-10'], pd.DataFrame.rolling,
            {'TMAX': 'max', 'TMIN': 'min', 'AWND': 'mean', 'PRCP':
```

```
'sum'},
            '3D').head()
```

datatype	TMAX	TMIN	AWND	PRCP
date				
2018-10-01	24.4	17.2	0.900000	0.0
2018-10-02	25.0	17.2	0.900000	17.5
2018-10-03	25.0	17.2	0.966667	17.5
2018-10-04	25.0	17.2	0.800000	18.5